

Can Regional Investment Promotion Policies Shield FDI from Uncertainty? Evidence from Brexit

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Abstract

Investment Promotion Agencies (IPAs) are a well-documented policy tool to attract and retain cross-border investment, with their effectiveness largely attributed to their role in reducing information asymmetries for prospective investors. Yet whether such a role holds in a global context increasingly characterised by uncertainty and trade fragmentation is a poorly understood and pressing question. This paper addresses this gap using the 2016–2019 Brexit negotiation period, exploiting heterogeneous pre-Brexit sector-targeting strategies across UK regional IPAs to test whether IPAs protected inward FDI during fundamental uncertainty, and if so how. Identification leverages the fact that Brexit was largely unanticipated, while IPA targeting strategies pre-dating 2015 (before Brexit was even contemplated) were motivated by regional industrial considerations rather than Brexit-proofing. I find that pre-2015 targeting strategies raised capital value by 18.4% and job creation by 20.6% in repeat FDI within Knowledge Intensive Services (KIS) sectors, with no effects for new FDI or manufacturing. This sectoral and investor-type pattern is informative about mechanism: effects accrue to established investors already embedded in regional ecosystems – for whom ecosystem benefits are realised rather than merely anticipated – and to sectors like KIS where Brexit uncertainty was less acute, where those benefits could plausibly offset higher (re)investment costs in an uncertainty context. Spatial and institutional heterogeneity reinforces this interpretation: effects concentrate outside South East England, where ecosystem assets are valuable but require active institutional brokering, and among Combined Authority IPAs with the capacity for sustained ecosystem coordination.

Keywords: Foreign Direct Investment; Investment Promotion Agencies; Brexit; Economic Uncertainty; Regional Development

JEL Codes: F21, F23, R11, R58

1 Introduction

Economic uncertainty sparked by shocks is known to negatively affect international trade and Foreign Direct Investment (FDI) (Dixit & Pindyck 1994). In an era of increasingly frequent disruptions – from geopolitical tensions and trade wars to pandemic-related supply chain fragmentation – attenuating the harmful effects of uncertainty on cross-border investment has become a pressing concern for policymakers worldwide (UNCTAD 2024).

A well-established literature points to institutions as key mitigating factors. By providing stability, reducing transaction costs, and offering operational support, institutional frameworks can reduce the ‘tax’ that uncertainty imposes on investment decisions (Buchanan et al. 2012, Mengitsu & Adhikary 2011, Globerman & Shapiro 2002). Yet institutional effectiveness is unlikely to be spatially uniform: regional economic structures, institutional capacities, and market access conditions shape both the severity of uncertainty shocks and institutions’ ability to moderate them.

Investment Promotion Agencies (IPAs) represent a particularly relevant institutional actor in this context. These public bodies aim to attract inward FDI and harness its developmental benefits (World Bank 2023). The literature identifies two broad channels through which IPAs may influence FDI. First, they reduce information asymmetries by clarifying regulatory requirements and administrative processes, lowering entry costs for multinational enterprises (MNEs) (Morisset & Andrews-Johnson 2004, Lim 2008, Harding & Javorcik 2011). Second, they foster regional ecosystem assets – specialised labour pools, supplier networks, research capabilities and industry clusters – which enhance firms’ embeddedness and operational performance (Beugelsdijk et al. 2010, Heilbron & Aranda Larrey 2020, Chidlow et al. 2015, Iammarino & McCann 2013).

While the information channel is well documented, robust empirical evidence on the ecosystem-building mechanism remains limited. More broadly, we lack clear empirical evidence on whether and how IPAs protect FDI during periods of fundamental uncertainty, which mechanism proves decisive, and how effects vary across space. Information provision addresses asymmetries where relevant knowledge exists but is costly to obtain. Fundamental uncertainty, by contrast, involves indeterminacy that cannot be resolved through information alone. In such settings, access to valuable regional ecosystems – which IPAs actively work to build and curate – may instead become the decisive factor. Yet whether this channel operates in practice, and under what institutional and geographic conditions, remains largely untested.

The UK’s unexpected Brexit vote in 2016 provides a compelling setting to examine this question. During the prolonged negotiation period (2016–2019), firms faced deep

uncertainty regarding future trade arrangements and single market access, posing a significant threat to UK investment (Bloom et al. 2019, Dhingra et al. 2016). Empirical evidence confirms substantially lower inward FDI following the referendum (Serwicka & Tamberi 2018). However, effects were heterogeneous across both sectors and space. Manufacturing sectors faced acute trade-related uncertainty due to potential tariff barriers and supply chain disruption, while Knowledge Intensive Services (KIS) sectors faced comparatively less immediate operational threats. At the same time, UK regions differed markedly in their pre-existing IPA sector-targeting strategies – strategies that in many cases were devised and implemented before Brexit was contemplated, and motivated primarily by regional industrial strengths rather than shock resilience. This paper exploits that cross-regional, cross-sectoral variation to examine whether IPAs protected inward FDI during fundamental uncertainty and through which mechanism.

I combine data on IPA sector-targeting strategies from the EU-funded MASSIVE survey (Crescenzi et al. 2018) with detailed FDI project data from Bureau van Dijk’s Orbis-crossborder database. A key feature of the identification strategy is the focus on strategies in place before 2015 – predating both the Brexit referendum and its announcement in the 2015 Conservative election manifesto – which ensures that targeting decisions plausibly reflect regional industrial considerations rather than Brexit-related responses. The richness of the data further allows a crucial distinction between new FDI by first-time investors and repeat FDI by established MNEs, whose investment decisions hinge on different factors and face different sensitivities to institutional support (Chidlow et al. 2015).

Using a Difference-in-Differences design over the 2010–2019 period, I find that pre-2015 IPA sector-targeting strategies increased inward FDI specifically in repeat investments by established MNEs and in KIS sectors. At the extensive margin, targeted regions were 4.1 percentage points more likely to receive repeat KIS FDI. At the intensive margin, they attracted 18.4% higher capital value and 20.6% higher job creation. No effects are detected for new FDI or manufacturing sectors. This pattern is consistent with ecosystem-building, rather than information provision, driving IPA effectiveness during fundamental uncertainty: established investors already embedded in regional ecosystems respond to IPA targeting, while new investors for whom information assistance would be most valuable – and manufacturing investors facing trade-related uncertainty that no amount of information can resolve – do not.

Effects exhibit marked spatial and institutional heterogeneity that further corroborates this interpretation. IPA effectiveness concentrates outside South East England – where ecosystem assets are valuable but less immediately accessible to external investors, making active institutional brokering most consequential – and among Combined Authority IPAs, whose greater statutory powers, devolved resources and coordination capacity

better equip them for the complex ecosystem-orchestration that uncertainty contexts demand (Pike et al. 2016, O'Brien & Pike 2015). Robustness checks confirm that results are not driven by flight-to-safety dynamics or pre-existing differential trends between targeted and non-targeted region-sectors.

These findings carry an important policy implication. Regional institutions are not rendered ineffective by global uncertainty shocks; rather, their role shifts. In stable times, IPAs attract investment primarily by reducing information barriers and coordination costs. During fundamental uncertainty, their value lies in the ecosystem relationships they have already built – connecting established investors to specialised regional assets that offset the costs of an uncertain environment. This protective function is strongest where institutional capacity is greatest and where regional ecosystems are valuable but not self-evidently accessible. By embedding firms within specialised regional ecosystems, IPAs thus shape differentiated territorial responses to transition-era disruptions and contribute to uneven but institutionally mediated patterns of regional resilience.

The remainder of the paper proceeds as follows. Section 2 develops the conceptual framework and hypotheses. Section 3 describes data and methodology. Section 4 presents results, and Section 5 discusses robustness checks. Additional results and robustness analyses are available upon request.

2 Conceptual Background & Literature

The Brexit vote in June 2016 mired the UK in a period of deep uncertainty regarding the nature of its future relationship with the EU (Bloom et al. 2019). Even if certain quarters had anticipated the possibility of a 'leave' vote ¹, the lack of clarity during the referendum campaign regarding exactly what Brexit entailed made it almost impossible for businesses to anticipate specific impacts and timelines (Dhingra & Sampson 2022). This kind of uncertainty has been well-documented to be particularly significant for foreign investors, effectively increasing the cost of investment (Dixit & Pindyck 1994, Buchanan et al. 2012, Mengitsu & Adhikary 2011).

Perhaps the biggest uncertainty derived from the spectre of losing access to the EU's single market for trade, services, capital and workers (Bailey & De Propris 2017). During negotiations, a wide range of potential outcomes were floated – from a 'soft' Brexit involving continued single market membership (akin to Norway) or customs union par-

¹In fact, the consensus among the literature is that Brexit was largely unexpected. Days before the referendum, pollsters put the chances of a 'leave' vote at under 30% (Bell, 2016), while studies of financial markets suggest little evidence these had priced in a leave vote (Davies & Studnicka, 2018). David Cameron, the Prime Minister responsible for calling the referendum, was so confident of a 'remain' win that he forbid government departments from making formal Brexit preparations.

ticipation (similar to Turkey), to a ‘hard’ Brexit with a limited free trade agreement (comparable to Canada) or even no deal at all – each with vastly different implications for trade frictions, regulatory alignment, and labour mobility. A second concern regarded potential regulatory and policy divergence in safety, quality and environmental standards that businesses must meet to sell into the EU Single Market (Li et al. 2019, Holweg 2019, Bailey & Rajic 2022). For certain MNE investors, the Brexit vote constituted a credible threat of higher future operating costs – whether through trade tariffs, labour costs or regulatory compliance – which significantly undermined the rationale for UK operations.

Given this fundamental uncertainty, the question arises: could regional institutions have played any role in moderating Brexit’s impact on FDI? There is growing recognition that territorial institutions – particularly at subnational or regional scale – may significantly influence MNE investment decisions by providing localised information, networks, and stability that buffer against broader uncertainties (OECD 2023, Crescenzi & Harman 2023, OECD 2026). Yet we have very little empirical evidence documenting this in action, either within the specific context of Brexit uncertainty or in other major shock contexts. In fact, the role of regional institutions in forging fruitful connections between MNEs and regions during periods of disruption has been termed a ‘black box’ in the literature (Yeung 2021). This section reviews what we know about how regional institutions, and Investment Promotion Agencies (IPAs) in particular, may influence FDI outcomes, before developing a conceptual framework and testable hypotheses for how these institutions might protect investment during fundamental uncertainty such as that created by Brexit.

2.1 Institutions, IPAs & FDI

Institutions are known to play a vital role in providing the guarantees necessary for economic transactions to occur (North 1990). In the context of FDI, they mitigate information asymmetries faced by MNEs entering new markets (or navigating shocks), reducing the operational risk effectively priced into cross-border investment decisions (Williamson 1975, 1985, Buchanan et al. 2012, Mengitsu & Adhikary 2011). Empirical research confirms that national institutional quality is a major determinant of FDI location (Bailey 2018). More recent work highlights that institutions operating at the regional scale also shape FDI location choices, particularly through spatial characteristics such as infrastructure, production linkages, and knowledge assets (Beugelsdijk et al. 2010). MNE location decisions are therefore not ‘space-blind’, but guided by the search for specific regional advantages (Dunning 2009).

Recognising this, regional institutions are increasingly mobilised to articulate these assets and enhance regional-sectoral ecosystems. Investment Promotion Agencies (IPAs)

represent a prominent example: public bodies tasked with attracting inward FDI and harnessing its developmental benefits (World Bank 2023). IPAs engage across the investment lifecycle, from initial marketing and lead generation to post-investment aftercare for retention and expansion support (Loewendahl 2018, Heilbron & Aranda Larrey 2020).

While this breadth of activity suggests multiple mechanisms, the literature has coalesced around two primary – though overlapping – roles. The first conceptualises IPAs as *information intermediaries* that reduce search costs and regulatory uncertainty for MNEs entering unfamiliar markets. By clarifying administrative processes and regulatory requirements, IPAs help overcome the “liability of foreignness” and lower informational barriers to investment (Morisset & Andrews-Johnson 2004, Lim 2008).

The second positions IPAs as *ecosystem orchestrators*, leveraging regional assets to connect MNEs with specialised suppliers, skilled labour, research institutions, and innovation networks (McCann & Mudambi 2005, Dunning 2009). Access to such assets influences both FDI attraction and subsidiary performance (Alfaro & Chen 2018), suggesting that institutions capable of facilitating ecosystem embeddedness may shape investment outcomes. In practice, IPAs foster supplier linkages, support collaboration with research organisations, and develop sector-specific skills initiatives, thereby reinforcing territorial embeddedness.

Empirical research confirms that IPAs increase FDI inflows (Harding & Javorcik 2011, Carballo et al. 2021), yet most studies emphasise the information-intermediary channel, particularly relevant for new investors facing entry-related uncertainty. In this view, IPAs reduce search costs and facilitate initial market entry. Emerging evidence, however, points toward a broader role. Crescenzi et al. (2021) find IPA effects concentrated in knowledge-intensive sectors, where access to localised capabilities and networks is likely to matter most. Forthcoming work by Alvarez-Vilanova *et al* suggests that IPA strategies are also effective in stimulating repeat investment and reducing divestment among already-established MNEs.

Together, these findings imply that IPA effectiveness may extend beyond entry to shaping post-entry expansion and retention decisions by firms embedded in regional economies. However, systematic empirical evidence isolating the ecosystem-orchestrator mechanism – and clarifying how its relevance differs between new and established investors – remains limited. In particular, we lack clear evidence on which channel proves more decisive under conditions of heightened uncertainty.

If ecosystem-building is indeed central in such contexts, its effectiveness is unlikely to be uniform. The ecosystem-orchestrator role is inherently coordination-intensive. Building the kinds of regional ecosystems that can shield against uncertainty requires sustained engagement across universities, industry associations, skills providers, and infrastruc-

ture bodies operating under distinct mandates and funding structures (Bachtler et al. 2019). Effective orchestration therefore depends on sufficient convening power and institutional authority to mobilise complementary resources (Sotarauta & Mustikkamäki 2010, Rodríguez-Pose & Garcilazo 2015). Institutional capacity thus becomes central to delivery.

In the UK, such capacity varies markedly across Combined Authorities, Local Enterprise Partnerships, and Upper Tier Local Authorities (Pike et al. 2016, O’Brien & Pike 2015). This heterogeneity, together with sectoral variation in Brexit exposure, provides a valuable setting to examine how institutional capacity moderates IPA effectiveness during fundamental uncertainty.

2.2 Conceptual Framework & Hypotheses

My conceptual framework starts from the understanding that IPAs may influence FDI decisions through two broad mechanisms: information sharing and ecosystem-orchestration. Under the information sharing mechanism, IPAs help MNEs by providing vital information and lowering information asymmetries in targeted sectors where they have built relationships and expertise, making (re)investment decisions less risky through better knowledge of regulatory requirements, administrative processes, and local market conditions. Under the ecosystem-orchestration mechanism, IPAs have an effect by taking action to build attractive regional ecosystems within which investors become carefully embedded – through aftercare services, fostering supplier linkages, developing sector-specific skills programmes, and facilitating connections with research institutions.

In the context of Brexit uncertainty, I hypothesise that ecosystem-orchestration becomes more decisive than information provision. Unlike stable settings where IPAs reduce identifiable information asymmetries, Brexit generated fundamental uncertainty that regional agencies could not resolve. Regional IPAs had little influence over national trade negotiations, and qualitative evidence suggests limited scope for providing materially new information during this period (Fuller 2021). Where uncertainty stems from indeterminacy rather than informational gaps, traditional assistance may be insufficient.

Instead, prior ecosystem-building activities may prove relatively decisive. Where IPAs had already fostered embeddedness in valuable regional ecosystems, established investors could continue to derive operational benefits that offset uncertainty costs. In such cases, repeat investment may remain attractive despite heightened national-level risk.

To investigate my overarching hypothesis – that IPA effectiveness during uncertainty operates primarily through ecosystem-orchestration rather than information-sharing – I define two testable hypotheses regarding the heterogeneous effects I may expect to find.

Because FDI of different types, sectors, and motivations is likely differentially influenced by Brexit uncertainty and more or less sensitive to different kinds of IPA activities, finding results in some categories over others can provide strong indication of the mechanism at play.

The first distinction I make is between new FDI and repeat FDI by established investors. If positive effects accrue to established investors rather than new FDI, this provides evidence in favour of the ecosystem-orchestration role, while the opposite would suggest information-sharing as the main channel. The intuition is twofold: first, established MNE investors considering repeat investment are more exposed to ecosystem-building activities because they already operate in the region and directly experience operational benefits, whereas new investors perceive these as potential rather than realised advantages. Second, established investors have already navigated market entry barriers, so their repeat FDI hinges relatively less on information provision and more on ecosystem embeddedness – access to skilled staff, suppliers, or R&D networks. Conversely, finding effects for new FDI would go against my main hypothesis and suggest that IPA roles in uncertainty contexts are much the same as in stable times.

The second distinction is between FDI in high Brexit uncertainty exposure sectors (manufacturing) and low exposure sectors (Knowledge Intensive Services). A larger effect for KIS FDI would provide further evidence that the operative channel is ecosystem-orchestration. Manufacturing sectors experienced particularly severe Brexit-related uncertainty, with concerns over future access to intricate, ‘just-in-time’ supply chains weighing heavily on operational and investment decisions (Bailey & De Propriis 2017, Bailey & Rajic 2022, Bloom et al. 2019). In such contexts, IPA impact may be limited: it matters little whether a region-sector has benefited from ecosystem-building activities if the broader locational advantage of UK access to the EU single market is at risk. Even established manufacturing investors may find that supply chain disruption outweighs the benefits of specialised regional ecosystems.

By contrast, KIS sectors faced different and often more manageable forms of uncertainty (Fuller 2021). While some KIS MNEs depended on EU market access, many threats could be addressed through less costly adjustments. The most significant concern – continued access to high-skill EU labour – was largely a longer-term rather than immediate operational issue. As a result, KIS sector MNEs contemplating investment may be more likely to proceed where ecosystem benefits – such as specialised R&D capabilities, collaboration with education and training institutions, or strong local policy networks – remain attractive. Conversely, if IPA strategies were found to affect manufacturing FDI, this would suggest that IPAs continue to operate, at least in part, through traditional information-assistance channels even under heightened uncertainty.

Table 1 summarises the conceptual framework and hypotheses discussed above. The following section outlines the data and empirical methodology that I take in order to explore these hypotheses.

Table 1: Summary of hypothesis and expected effects, vs. the ‘counter-hypothesis’

Hypothesis	Counter Hypothesis
H1: In times of uncertainty, IPAs’ role in growing FDI is limited to the ecosystem-building channel	C1: In times of uncertainty, IPAs’ role in growing FDI continues to operate through channel of information assistance
Hypothesised effects	Hypothesised effects (counter)
Effect of IPAs concentrates on FDI types deriving relatively greater benefit from ecosystem factors, specifically:	Effect of IPAs is predominantly on FDI types deriving relatively greater benefit from information assistance:
H1.1: Repeat FDI	C1.1: New FDI
H1.2: KIS sector FDI	C1.2: Manufacturing FDI

3 Data & Methodology

To explore these hypotheses, I combine information on the sector-targeting strategies in place by IPAs with geo-located, UK-bound inward FDI from July 2010 to July 2019. The result is a panel dataset of region-sectors in the UK with ‘treatment’ dummies identifying those that were exposed to an IPA targeting strategy in the pre-Brexit period, and outcome data in terms of inward FDI (both new and repeat investment) both in the pre-Brexit vote (July 2010–June 2016) and post-Brexit vote (July 2016–June 2019) period. This section describes the data sources and presents descriptive patterns that motivate my empirical approach.

3.1 Data on Inward FDI into Great Britain

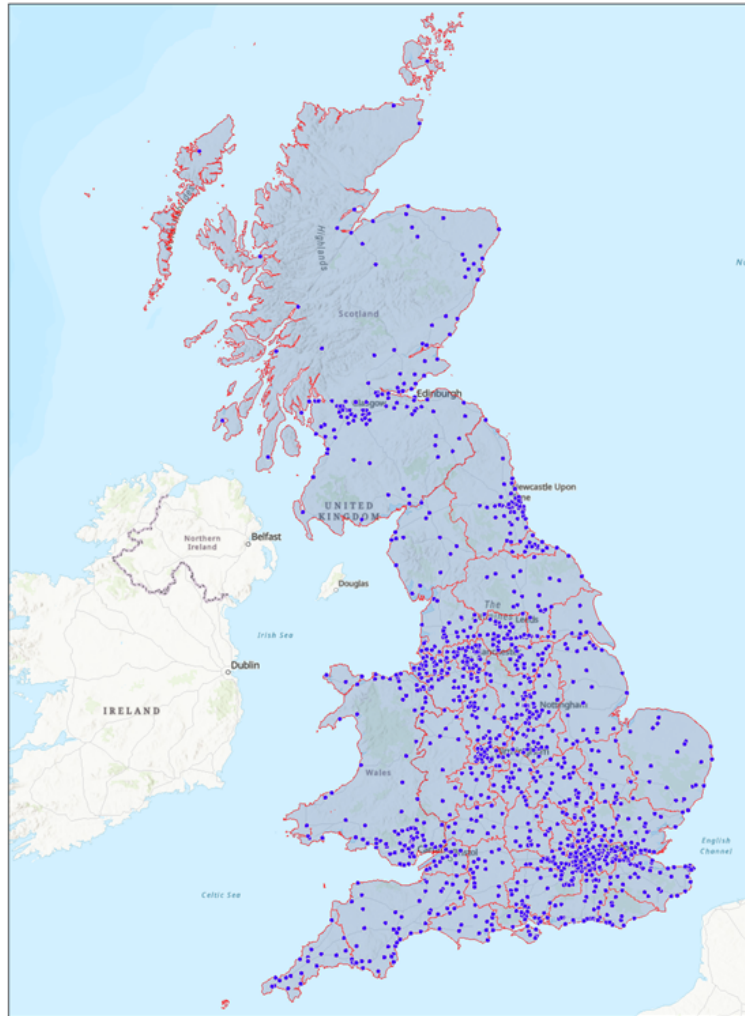


Figure 1: **Distribution of FDI projects into Great Britain, 2010-2019.**

Source: Orbis-crossborder (Bureau van Dijk / Moody's Analytics 2024).

Data on inward FDI are sourced from the Orbis-crossborder database by Bureau van Dijk (BvD) / Moody's Analytics (Bureau van Dijk / Moody's Analytics 2024). This provides geo-located information on every new FDI project/expansion made in a UK region going as far back as 2003, including capital value, jobs created and the project description, by NACE sector. In the empirical literature, a frequently-used inward FDI dataset is the fDiMarkets database by the Financial Times. However, I opt for the Orbis-crossborder dataset, as it contains systematic information on key variables such as the FDI motivation or the project description, which enables better exploration of treatment effect heterogeneity. The provision of NACE sector that each FDI project takes place in, which is present in the Orbis-crossborder data but not fDiMarkets², is also useful to more

²fDi Markets instead sorts FDI projects into their own sectoral classification, which would require applying a crosswalk onto the NACE classification.

neatly identify investments in Knowledge-Intensive Services (KIS) and manufacturing sectors, which following the discussion in section 2 constitute a key distinction for my analysis.

Table 2: Basic descriptives on inward FDI projects in Orbis-crossborder database for Great Britain, 2010–2019

FDI Project Type	Num. Projects	FDI Cap. Value (USD M)	FDI Jobs
New	9,013	198,702	503,539
Repeat	3,422	114,404	256,205
Total	12,435	313,106	759,744

Source: Own elaboration with data from Orbis-crossborder by BvD/Moody’s Analytics.

Given that the Brexit withdrawal agreement was signed in January 2020, plus the fact that the COVID-19 pandemic that closely followed caused a whole uncertainty shock of its own, I restrict the dataset to investments occurring from July 2010 through to June 2019. Moreover, I exclude Northern Ireland from my study, as its unique situation regarding the Irish border – which needed to remain open under the Good Friday Agreement – likely rendered it subject to a different nature and severity of uncertainty compared to Great Britain. Basic descriptives for FDI projects, in terms of number of events, FDI value and job creation, are shown in Table 2.

To contextualise the Brexit uncertainty period within broader European trends, Figure 2 plots the evolution of inward FDI into the UK relative to the five largest EU27 economies. While FDI into comparator countries increased markedly after mid-2016, UK-bound FDI remained largely flat. This divergence is consistent with existing evidence that the UK experienced a Brexit-related decline in inward investment relative to a synthetic counterfactual (Serwicka & Tamberi 2018), and supports the interpretation of Brexit as a UK-specific uncertainty shock rather than part of a broader European slowdown.

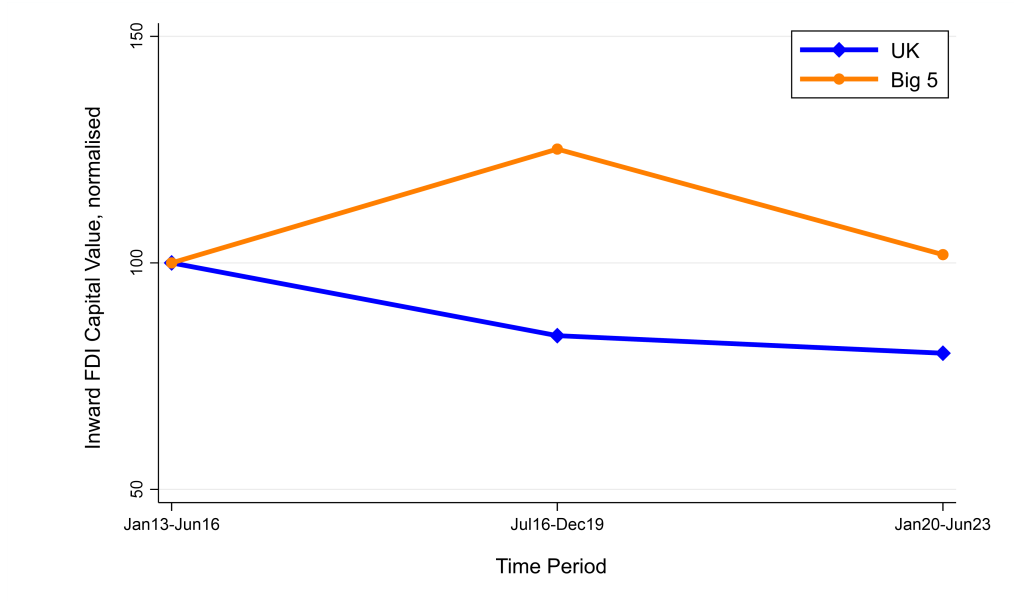


Figure 2: **Total value of inward FDI: UK vs. Big 5 EU27 economies (normalised).** Big 5 refers to Germany, France, Italy, Spain and Poland.

Source: Own elaboration using data from Orbis-crossborder (Bureau van Dijk / Moody's Analytics 2024).

3.2 Data on IPA Targeting Strategies

Data on the sector-specific targeting strategies of UK IPAs were collected through the 2018 MASSIVE survey of IPAs in Europe (Crescenzi et al. 2021). This covers 42 IPAs, 40 of which are English, sub-national IPAs while the remaining two are the national IPAs of the two devolved administrations in Great Britain (Scotland and Wales). These are shown in Figure 3, which distinguishes IPAs by institutional form as at 2018, and listed in the Appendix.

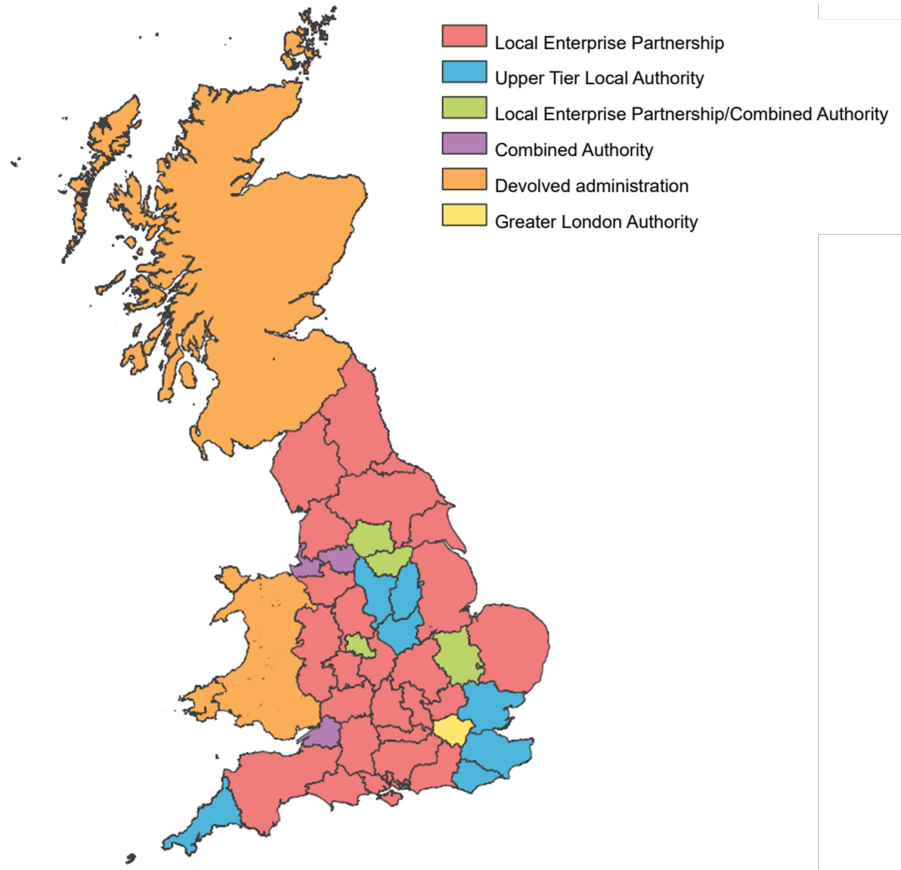


Figure 3: **Sample of IPAs in Great Britain, by institution type.**
Source: Own elaboration, MASSIVE Survey (2018).

3.2.1 Regional & Sectoral Variation in IPA Targeting Strategies

Of the 42 surveyed IPAs, 34 had sector-targeting strategies at any point in time and 22 had sector-targeting strategies in place before 2015 (and which can therefore be reasonably assumed to have been implemented independently of Brexit and its potential effects). My treatment group consists of the 218 region-sectors which were targeted by the corresponding IPA before 2015, to reflect the fact that Brexit was ‘announced’ in the 2015 Election Conservative Manifesto (published in March 2015). My control group comprises the remaining 622 region-sectors, excluding the few cases where an IPA started targeting a particular sector after 2015 (as these targeting decisions may well have been motivated by Brexit).

Table 3 shows that the treatment and control groups are broadly comparable in terms of regional and sectoral composition, though treated region-sectors are somewhat more likely to be in manufacturing (46.8% vs. 44.4%) and KIS sectors (25.2% vs. 18.2%).

Table 3: Composition of treatment and control groups

Variable	Treatment group	Control group	All
Num. region-sectors	218	622	840
Num. regions represented	24	41	42
<i>Regional composition:</i>			
Share of regions in South East England ^a	26.2%	26.1%	26.2%
Share of Combined Authorities	20.2%	15.5%	16.7%
Share of Local Enterprise Partnerships	39.0%	66.8%	59.5%
Share of County Councils	28.4%	12.6%	16.7%
Share Devolved Administrations ^a	12.4%	5.1%	7.1%
<i>Sectoral composition:</i>			
Num. sectors represented	20	20	20
Share of primary sectors	5.5%	10.6%	10.0%
Share of knowledge-intensive service sectors	25.2%	18.2%	20.0%
Share of other service sectors	18.3%	20.6%	20.0%
Share of manufacturing sectors	46.8%	44.4%	45.0%
Share of energy/utilities sectors	4.1%	5.3%	5.0%

^aIncluding Greater London. Source: MASSIVE Survey (2018).

Figure 4 shows the sectoral distribution of IPA targeting strategies, distinguishing between strategies implemented at any point and those in place before 2015. There is considerable heterogeneity across sectors: while some sectors like food products and biotechnology were often targeted, others saw limited or late adoption of targeting strategies. This variation in regional strategies provides an observable dimension through which to examine whether and how regional institutions may have moderated the negative effects of Brexit uncertainty on inward FDI.

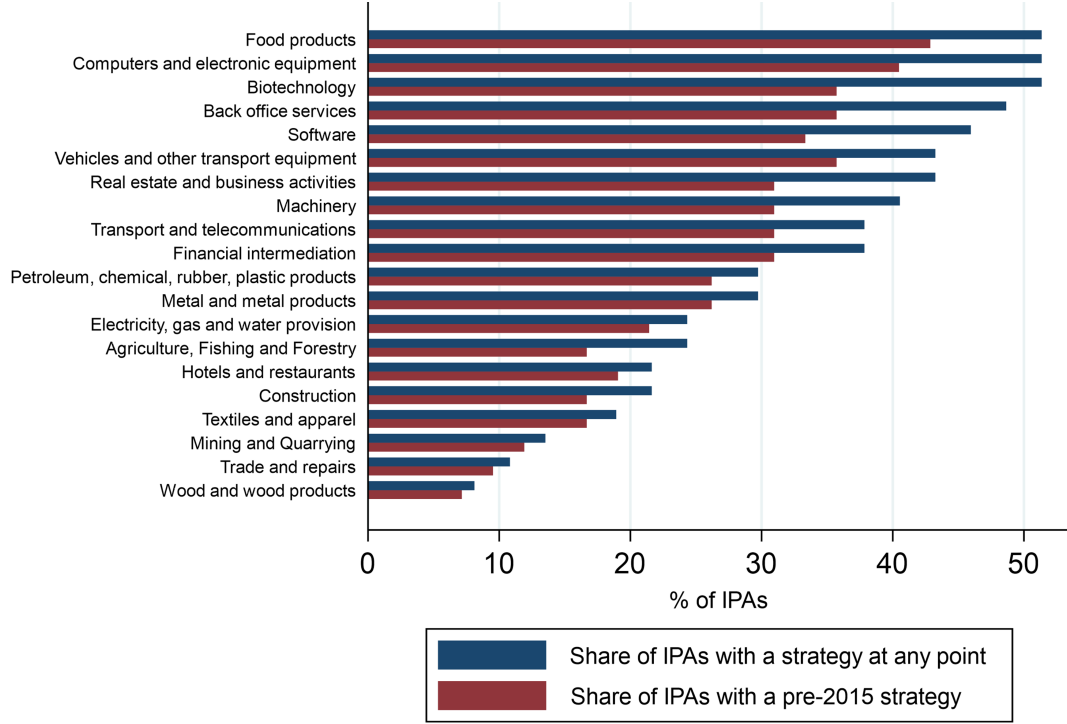


Figure 4: **Share of regional IPAs with sector-based strategies by sector.**

Source: MASSIVE Survey (2018).

3.3 Empirical Approach

My empirical approach builds on Harding & Javorcik (2011) and Crescenzi et al. (2021), exploiting variation in sector-specific IPA targeting strategies to gauge causal impacts on inward FDI. Identifying such effects is empirically challenging because sector targeting is not random, but rather informed by pre-existing regional strengths or industrial objectives (Loewendahl 2001, 2018). Additionally, regions or sectors may be affected by contemporaneous shocks that correlate with treatment designation and confound results.

To mitigate this, past studies control for confounding variation through time, region and sector fixed effects, and employ staggered difference-in-difference specifications to verify pre-treatment parallel trends (Angrist & Pischke 2008). My approach builds on this work but deviates in two key ways. First, my treatment is the interaction of region-sector exposure to an IPA strategy and being in the post-Brexit period, rather than a time-varying treatment dummy. Since treatment ‘switches on’ simultaneously for all treated units, I estimate a standard Two Way Fixed Effects (TWFE) model. Second, I distinguish from the outset between new FDI and repeat FDI, in reflection of the fact that these face different obstacles and must therefore be considered separately.

Specifically, I estimate the following equations, separately for new and repeat FDI:

$$Y_{rst} = \beta_0 + \beta_1(Targeted_{rs} \times Post_Brexit_t) + \alpha_{rs} + \delta_{rt} + \gamma_{st} + \varepsilon_{rst} \quad (1)$$

Where Y_{rst} denotes either new or repeat FDI into region r , sector s , time-period t , measured at both extensive (any FDI) and intensive margins (IHS of capital value, jobs created, and project count).³ Time periods are defined as 12-month intervals from July 2010 through June 2019, yielding nine periods.⁴ Knowledge-intensive service (KIS) sectors follow Eurostat’s classification of NACE codes; manufacturing sectors comprise all NACE segment C codes.

$Targeted_{rs}$ equals one if region r targeted sector s before 2015, as gauged from the MASSIVE survey (Crescenzi et al. 2021). This cutoff excludes strategies potentially motivated by Brexit fears following the April 2015 Conservative manifesto pledge, ensuring pre-treatment targeting reflects regional strengths rather than Brexit-related considerations. Region-sectors where IPA strategies were implemented in 2015 or later are excluded from the sample. $Post_Brexit_t$ denotes periods after the Brexit referendum (from July 2016). The coefficient of interest, β_1 , captures the average change in Y_{rst} during the post-Brexit period for targeted region-sectors relative to untargeted ones. α_{rs} , δ_{rt} , and γ_{st} denote region-sector, region-time, and sector-time fixed effects, and ε_{rst} is a heteroskedasticity-robust error term clustered at the region-sector level.

To evaluate dynamic effects and assess pre-treatment parallel trends, I complement equation (1) with an event-study specification that replaces the post-treatment indicator with period-specific interactions:

$$Y_{rst} = \beta_0 + \sum_{k \neq -1} \beta_k(Targeted_{rs} \times Period_k) + \alpha_{rs} + \delta_{rt} + \gamma_{st} + \varepsilon_{rst} \quad (2)$$

Here, β_k coefficients capture the difference between treated and control region-sectors in each period k relative to the final pre-treatment period (15-16), which serves as the base. Pre-treatment β_k coefficients that are close to zero and statistically insignificant support the parallel trends assumption underlying identification.

For the main analysis, I exclude Greater London, Scotland, and Wales. These IPAs operate at fundamentally different scales than English regional IPAs: Greater London and the devolved administrations command substantially greater resources, institutional authority, and policy autonomy. Including them in the baseline specifications produces

³The IHS transformation is preferred over the natural logarithm because it accommodates zero values, which are common in my data. The transformation is defined as $IHS(x) = \ln(x + \sqrt{x^2 + 1})$, which approximates the natural logarithm for large values while retaining zero-valued observations.

⁴The choice of 12-month periods reflects data considerations around sufficient FDI activity per region-sector-period cell. Monthly or quarterly aggregation would result in many zero-valued cells, particularly for smaller regions or less popular sectors.

unstable estimates that obscure patterns for typical regional agencies. Results including these IPAs are presented in the Appendix and show consistent patterns.

Exploring my hypotheses about IPAs' role during heightened uncertainty requires moving beyond average treatment effects. I therefore estimate equations (1) and (2) for subsets of the data, examining heterogeneous effects by FDI sector and region characteristics. As discussed in Section 2.2, if ecosystem-building dominates, effects should concentrate among repeat FDI and KIS sectors, outside of South East England where ecosystem infrastructure is less immediately accessible, and among Combined Authority IPAs with the institutional capacity for complex ecosystem coordination. These heterogeneous effects are explored through subsample analysis.

3.3.1 Additional Empirical Challenges and Robustness Tests

While the baseline specifications incorporate rich fixed effects and dynamic treatment models to assess pre-treatment trends, three additional identification concerns warrant explicit discussion. Full results for each are presented in Section 5.

Normal-times IPA effectiveness. IPA targeting is documented to be effective during stable economic periods (Harding & Javorcik 2011, Crescenzi et al. 2021). If IPAs were already attracting additional FDI before Brexit, positive pre-trends in equation (2) could reflect legitimate baseline effectiveness rather than violations of parallel trends. The key question in this paper is therefore not whether IPAs work in normal times, but whether Brexit uncertainty altered the magnitude, sectoral distribution, or mechanism of their effectiveness. To address this, I conduct a complementary event study focusing exclusively on the pre-Brexit period (2010–2016), tracing FDI outcomes around the introduction of targeting strategies during stable times. This specification follows the structure of equation (2) but is estimated on the pre-referendum subsample with staggered treatment timing, using both standard dynamic TWFE and the Sun & Abraham (2021) estimator to account for potential biases in staggered adoption settings.

Flight-to-safety dynamics. During periods of heightened uncertainty, investors may disproportionately favour region-sectors with historically strong performance, independent of institutional targeting. If targeted region-sectors systematically coincide with structurally strong performers, estimated effects could capture resilience driven by pre-existing advantages rather than IPA strategy. To address this, I construct a composite pre-Brexit strength score capturing both the quality and magnitude of each region-sector's historical FDI specialisation:

$$StrengthScore_{rs} = \left(\frac{\sum_{t < 2015} FDI_{rst}^{HighQuality}}{\sum_{t < 2015} FDI_{rst}^{All}} \right) \times \left(\frac{\sum_{t < 2015} FDI_{rst}^{HighQuality}}{\sum_{r, t < 2015} FDI_{st}^{HighQuality}} \right) \quad (3)$$

The first term measures the share of a region-sector’s FDI that is high-quality (based on business functions indicating headquarters, R&D, or innovation activities). The second term measures the region-sector’s share of all high-quality FDI in its sector nationally, capturing its importance as a quality investment destination. I define $Leading_{rs} = \mathbb{1}(StrengthScore_{rs} > 0.1)$ and augment equation (2) by interacting $Leading_{rs}$ with time period dummies, alongside the $Targeted_{rs} \times Period_k$ interactions. This allows me to distinguish FDI resilience attributable to historically strong region-sectors from the incremental effect of IPA targeting. Results are robust to alternative thresholds (0.05, 0.20).

Pre-existing differential trends. While the event study in equation (2) assesses parallel trends graphically and statistically, I further implement placebo-style specifications assigning false treatment dates in the pre-Brexit period to verify that no spurious treatment effects emerge when no genuine uncertainty shock occurred.

4 Results

This section outlines the results of the regressions specified to verify the hypotheses outlined in section 2.3. For each of my two main outcomes – capital value and number of projects – I show results for three variants of my specification: first, a ‘simple’ specification that includes just 12-month period and region-sector fixed effects; second, a specification that adds region-by-time and sector-by-time fixed effects to absorb changes in my outcome explained by region-specific or sector-specific shocks; and third, a specification that additionally controls for the interaction between pre-Brexit FDI growth rates and the post-Brexit period, addressing the concern that IPA targeting may have correlated with pre-existing growth trajectories. Results for another two outcomes (number of FDI projects and job creation) are presented in the Appendix.

Table 4: Impact of pre-Brexit IPA targeting on inward FDI into English regions (excl. Greater London), 2010–2019, new vs. repeat investments

	IHS cap. value			Dummy FDI		
	(1)	(2)	(3)	(4)	(5)	(6)
NEW FDI						
Pre-2015 IPA targeting × Post-Brexit	0.001 (0.096)	-0.010 (0.126)	-0.014 (0.125)	0.001 (0.021)	-0.016 (0.030)	-0.017 (0.030)
Observations	6,900	6,900	6,900	6,900	6,900	6,900
R-squared	0.411	0.497	0.497	0.430	0.506	0.506
REPEAT FDI						
Pre-2015 IPA targeting × Post-Brexit	0.308*** (0.097)	0.325** (0.140)	0.323** (0.141)	0.056** (0.024)	0.057 (0.036)	0.057 (0.036)
Observations	6,900	6,900	6,900	6,900	6,900	6,900
R-squared	0.332	0.412	0.412	0.343	0.427	0.427
Region-Sector FE	Yes	Yes	Yes	Yes	Yes	Yes
Sector×Time FE	No	Yes	Yes	No	Yes	Yes
Region×Time FE	No	Yes	Yes	No	Yes	Yes
Pre-Brexit Growth × Post	No	No	Yes	No	No	Yes

*** p<0.01, ** p<0.05, * p<0.1. *Standard errors are clustered at the region-sector level. New FDI includes Orbis-crossborder projects categorized as “Greenfield” or “New foreign location”. Repeat FDI includes projects categorized as “Expansions”, “Co-locations” and “Re-locations” under variable “Project type”. Dummy FDI indicates whether any FDI project occurred in a given region-sector-time cell (extensive margin). Time periods are 12-month intervals from July 1st to June 30th, beginning on July 1st 2010 and ending in June 30th 2019 (giving 9 time periods). Unit of analysis are region-sector-12month cells. Greater London, Scotland, and Wales are excluded from all regressions, given that these IPAs operate at a fundamentally different scale than regional IPAs, and their inclusion produces unstable estimates that obscure patterns for typical regional agencies. That said, results including these IPAs show similar patterns, and are available upon request. Region-sectors where an IPA strategy was put in place in 2015 or later, and which can be more reasonably assumed to have been influenced by the announcement of the Brexit referendum within the 2015 Conservative election manifesto, are dropped from all regressions.*

Table 4 shows results for my full sample of region-sectors. It can be seen how IPA targeting appears to have no effect on new inward FDI into those region-sectors that were targeted. For repeat FDI, however, there is in fact some evidence that such strategies led to a positive increase in FDI during the Brexit negotiation period, particularly when measured at the intensive margin (FDI capital value).

To dig further into these results for repeat FDI, and in line with the theoretical distinction between information-provision and ecosystem-building mechanisms, Table 5 distinguishes between repeat FDI in KIS and manufacturing sectors. As discussed in section 2.2, each faced different threats deriving from Brexit uncertainty, and may therefore exhibit different reactions to treatment.

Table 5: Impact of pre-Brexit IPA targeting on inward repeat FDI into English regions (excl. Greater London), 2010–2019, manufacturing vs. KIS sector

	IHS cap. value			Dummy FDI		
	(1)	(2)	(3)	(4)	(5)	(6)
MANUFACTURING FDI						
Pre-2015 IPA targeting × Post-Brexit	0.158**	0.124	0.122	0.038**	0.022	0.022
	(0.073)	(0.112)	(0.112)	(0.016)	(0.024)	(0.025)
Observations	6,900	6,900	6,900	6,900	6,900	6,900
R-squared	0.342	0.399	0.399	0.325	0.386	0.386
KNOWLEDGE-INTENSIVE SERVICES FDI						
Pre-2015 IPA targeting × Post-Brexit	0.151***	0.171***	0.169***	0.038**	0.041**	0.041**
	(0.052)	(0.064)	(0.064)	(0.015)	(0.020)	(0.020)
Observations	6,900	6,900	6,900	6,900	6,900	6,900
R-squared	0.474	0.547	0.547	0.465	0.538	0.538
Region-Sector FE	Yes	Yes	Yes	Yes	Yes	Yes
Sector×Time FE	No	Yes	Yes	No	Yes	Yes
Region×Time FE	No	Yes	Yes	No	Yes	Yes
Pre-Brexit Growth × Post	No	No	Yes	No	No	Yes

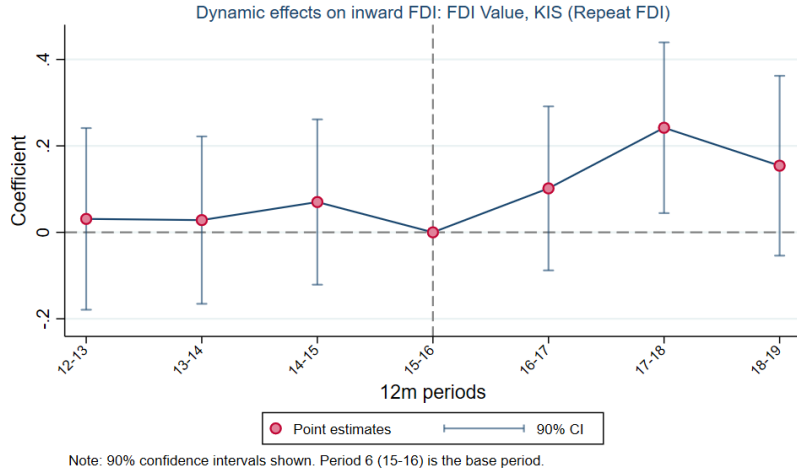
*** p<0.01, ** p<0.05, * p<0.1. Standard errors are clustered at the region-sector level. Each outcome refers to FDI projects listed in Orbis-crossborder as “Expansions”, “Co-locations” and “Re-locations” under variable “Project type”. Dummy FDI indicates whether any FDI project occurred in a given region-sector-time cell (extensive margin). Time periods are 12-month intervals from July 1st to June 30th, beginning on July 1st 2010 and ending in June 30th 2019 (giving 9 time periods). Unit of analysis are region-sector-12month cells. KIS sectors identified in line with Eurostat definition of KIS NACE sectors, as detailed here: [https://ec.europa.eu/eurostat/statistics-explained/index.php?title=Glossary:Knowledge-intensive_services_\(KIS\)](https://ec.europa.eu/eurostat/statistics-explained/index.php?title=Glossary:Knowledge-intensive_services_(KIS)). Manufacturing sector FDI are all projects falling under NACE code C. Greater London, Scotland, and Wales are excluded from all regressions, given that these IPAs operate at a fundamentally different scale than regional IPAs, and their inclusion produces unstable estimates that obscure patterns for typical regional agencies. That said, results including these IPAs show similar patterns, and are available upon request. Region-sectors where an IPA strategy was put in place in 2015 or later, and which can be more reasonably assumed to have been influenced by the announcement of the Brexit referendum within the 2015 Conservative election manifesto, are dropped from all regressions.

It can be seen how IPA strategies were effective in protecting repeat FDI where these applied to KIS sectors. For manufacturing sectors – where trade-related Brexit uncertainty was particularly acute – we find no robust evidence of IPA targeting effectiveness. While the simple specification (column 1) suggests a positive effect, this disappears when we control for region-specific and sector-specific time trends (columns 2-3), indicating that the initial result likely reflected confounding shocks rather than genuine treatment effects.

For KIS sectors, IPA targeting strategies implemented pre-2015 are associated with substantially larger repeat FDI investments during the Brexit uncertainty period, with effects concentrated on the intensive margin. Region-sectors where such strategies were in place received investments that were, on average, 18.4% larger in capital value com-

pared to non-targeted region-sectors.⁵ For my alternative intensive-margin outcome, IHS-transformed job creation (shown in the Appendix), a statistically-significant effect of similar magnitude is found, with treated region-sectors seeing 20.6% more job creation. By contrast, effects on the extensive margin are weaker and less robust: while (as shown in Table 5) targeted region-sectors were 4.1 percentage points more likely to receive any repeat KIS FDI, the effect on the count of FDI projects (shown in the Appendix) is weaker and does not survive estimation via Poisson Pseudo-Maximum Likelihood – the appropriate estimator for count data – indicating sensitivity to distributional assumptions. Taken together, this pattern suggests that IPA targeting primarily influenced the scale of existing investors’ commitments rather than determining whether investment occurred at all.

Figure 5: Dynamic treatment effects on KIS repeat FDI (capital value)



Note: 90% confidence intervals shown. Period 6 (15-16) is the base period. Periods 10-11 and 11-12 omitted for clarity. Standard errors clustered at the region-sector level. Dynamic effects estimated using the preferred specification (region-sector, region×time and sector×time fixed effects, controlling for pre-Brexit FDI growth interacted with the post-Brexit indicator). Outcome is IHS-transformed repeat KIS FDI capital value. Region-sectors where an IPA strategy was put in place in 2015 or later are excluded.

Figure 5 presents dynamic treatment estimates for capital value, the outcome on which effects are most pronounced. Pre-treatment coefficients are statistically indistinguishable from zero throughout, consistent with parallel pre-trends prior to the Brexit referendum. Post-Brexit, treatment effects emerge and strengthen across the three post-treatment periods, consistent with IPAs providing sustained protection against uncertainty shocks rather than a one-off response. The equivalent event study for job creation – the other intensive-margin outcome – is shown in Figure A1 and follows a closely similar pattern. When including Greater London, Scotland and Wales (see Figure A2), which as discussed

⁵Percentage changes are calculated as $(e^{\beta} - 1) \times 100$ where β is the estimated coefficient from the preferred specification (column 3) using IHS-transformed outcomes.

are not considered fully comparable to English IPAs in terms of resourcing and capabilities, dynamic effects follow a similar pattern — if anything, the post-treatment effect becomes stronger.

Dynamic estimates for manufacturing repeat FDI (reported in the Appendix) confirm the absence of robust treatment effects, with post-treatment coefficients statistically indistinguishable from zero. Notably, pre-treatment coefficients are also positive and of similar magnitude, suggesting no structural break at the onset of Brexit uncertainty – consistent with IPAs having little differential impact on manufacturing FDI regardless of period.

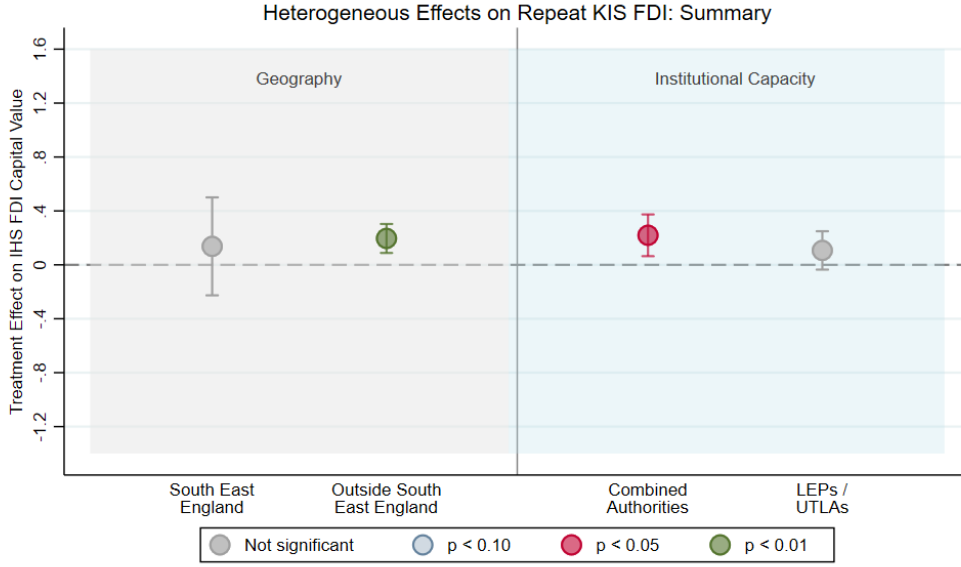
This pattern raises a natural question: do the positive effects observed for KIS repeat FDI truly reflect a distinct uncertainty-period mechanism, or do they simply represent a continuation of IPA effectiveness already present in normal times? I formally address this in Section 5 through pre-Brexit event studies that isolate targeting effects during stable periods. As shown there, IPA targeting does not exhibit statistically significant associations with repeat FDI prior to 2016, reinforcing the interpretation that the KIS repeat FDI effects documented here emerge specifically during the Brexit uncertainty period.

4.1 Heterogeneous Effects

In order to further characterise the channel through which IPA targeting operates to protect inward FDI during times of uncertainty, in this section I further distil results by geography and institutional capacity. If stronger effects accrue to regions that are traditionally less economically dynamic, or to regions where IPAs have greater institutional capacity, then this may provide reinforcing evidence that the mechanism in play is more associated with ecosystem-building than with information-sharing.

Figure 6 summarises the key heterogeneity results. Two dimensions are shown: geographic location (left panel) and institutional capacity (right panel). Colour coding indicates statistical significance, with green denoting significance at the 1% level, red at 5%, and grey indicating insignificance.

Figure 6: Heterogeneous Effects on Repeat KIS FDI: Summary



Note: Each point shows the coefficient on `post_treat` from a separate regression estimated on the indicated subsample, using the preferred specification (region-sector, region×time and sector×time fixed effects, controlling for pre-Brexit FDI growth interacted with the post-Brexit indicator). Outcome is IHS-transformed repeat KIS FDI capital value. 90% confidence intervals shown, with standard errors clustered at the region-sector level. Greater London, Scotland and Wales are excluded throughout. South East England is defined as belonging to a UK NUTS1 region contiguous with Greater London. Combined Authority IPAs include: Greater Manchester CA, West Midlands CA, Cambridgeshire & Peterborough CA, Tees Valley CA, West of England CA, Sheffield City Region, Leeds City Region, Liverpool City Region and North East Combined Authority. Region-sectors where an IPA strategy was put in place in 2015 or later are excluded.

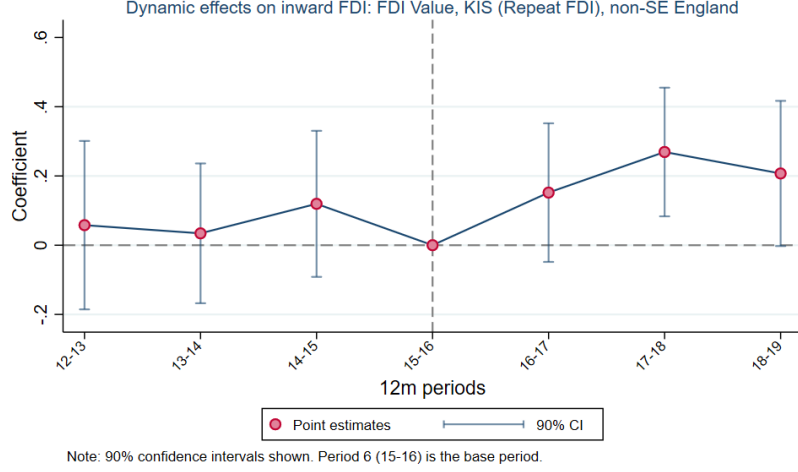
4.1.1 Geographic Heterogeneity

The geographic results strongly support the ecosystem-building prediction. IPA targeting shows a positive and highly significant effect outside South East England, while no significant effect is found within the South East. This pattern is consistent with ecosystem-building mechanisms adding greatest value precisely where regional infrastructure must be actively constructed rather than where it already exists (Rodríguez-Pose & Garcilazo 2015, Beugelsdijk et al. 2010). Within the economically prosperous South East core – where firms already have ready access to dense networks, specialised labour, and research institutions – IPA-facilitated ecosystem access may simply be redundant. Outside this core, by contrast, regional assets are valuable but less immediately accessible to external investors, making the IPA’s role in brokering these connections more decisive.

Figure 7 presents the dynamic treatment estimates for Outside South East England. Pre-treatment coefficients are statistically indistinguishable from zero throughout, consistent with parallel pre-trends. Post-Brexit, treatment effects emerge and strengthen persistently across all three post-treatment periods, confirming that IPA effectiveness

switched on during the Brexit uncertainty period specifically in regions where ecosystem-building activities are most valuable.

Figure 7: Dynamic Treatment Effects on KIS Repeat FDI – Outside South East England



Note: 90% confidence intervals shown. Period 6 (15-16) is the base period. Periods 10-11 and 11-12 omitted for clarity. Standard errors clustered at the region-sector level. Dynamic effects estimated using the preferred specification (region-sector, region $\times$ time and sector $\times$ time fixed effects, controlling for pre-Brexit FDI growth interacted with the post-Brexit indicator). Outcome is IHS-transformed repeat KIS FDI capital value. Greater London, Scotland and Wales are excluded throughout. South East England is defined as belonging to a UK NUTS1 region contiguous with Greater London.

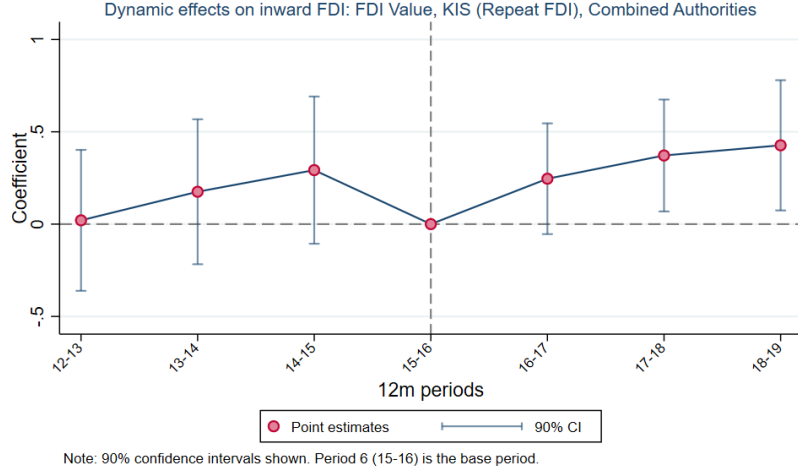
4.1.2 Institutional Capacity

The institutional capacity results are equally informative. Combined Authority IPAs – statutory bodies with devolved powers, dedicated investment promotion staff, and substantially greater resources than Local Enterprise Partnerships or Upper Tier Local Authorities – show a positive and significant effect, while the latter show no significant impact. This stark contrast suggests that ecosystem-building during uncertainty requires the kind of institutional capacity that enables sustained coordination across universities, industry associations, skills providers and infrastructure bodies – precisely the convening power and formal authority that Combined Authorities possess and that fragmented LEPs and UTLAs lack (Pike et al. 2016, O’Brien & Pike 2015).

Figure 8 presents dynamic treatment estimates for Combined Authority regions. Unlike the Outside South East results, these exhibit positive pre-treatment coefficients – a pattern worth acknowledging directly. This pre-existing upward trend is consistent with Combined Authorities being precisely the kind of high-capacity institutions whose ecosystem-building activities generate FDI returns even in normal times. What the post-Brexit period reveals is an accentuation of this pattern under uncertainty: Combined Authorities’ pre-established ecosystem relationships proved particularly decisive when

broader uncertainty threatened investment, as the operational benefits they had built up over time helped offset the costs of an uncertain investment environment. The post-treatment coefficients are clearly positive, larger in magnitude than the pre-treatment fluctuations, and consistent in sign across all three post-Brexit periods.

Figure 8: Dynamic Treatment Effects on KIS Repeat FDI – Combined Authorities



Note: 90% confidence intervals shown. Period 6 (15-16) is the base period. Periods 10-11 and 11-12 omitted for clarity. Standard errors clustered at the region-sector level. Dynamic effects estimated using the preferred specification (region-sector, region $\times$ time and sector $\times$ time fixed effects, controlling for pre-Brexit FDI growth interacted with the post-Brexit indicator). Outcome is IHS-transformed repeat KIS FDI capital value. Combined Authority IPAs include: Greater Manchester CA, West Midlands CA, Cambridgeshire & Peterborough CA, Tees Valley CA, West of England CA, Sheffield City Region, Leeds City Region, Liverpool City Region and North East Combined Authority. Region-sectors where an IPA strategy was put in place in 2015 or later are excluded.

5 Robustness Checks

This section reports three sets of robustness results that directly correspond to the identification concerns outlined in Section 3.1.1: (i) whether IPA effectiveness during Brexit uncertainty simply reflects pre-existing normal-times effectiveness, (ii) whether flight-to-safety dynamics bias estimates upward, and (iii) whether results are driven by pre-existing differential trends rather than genuine responses to the Brexit shock.

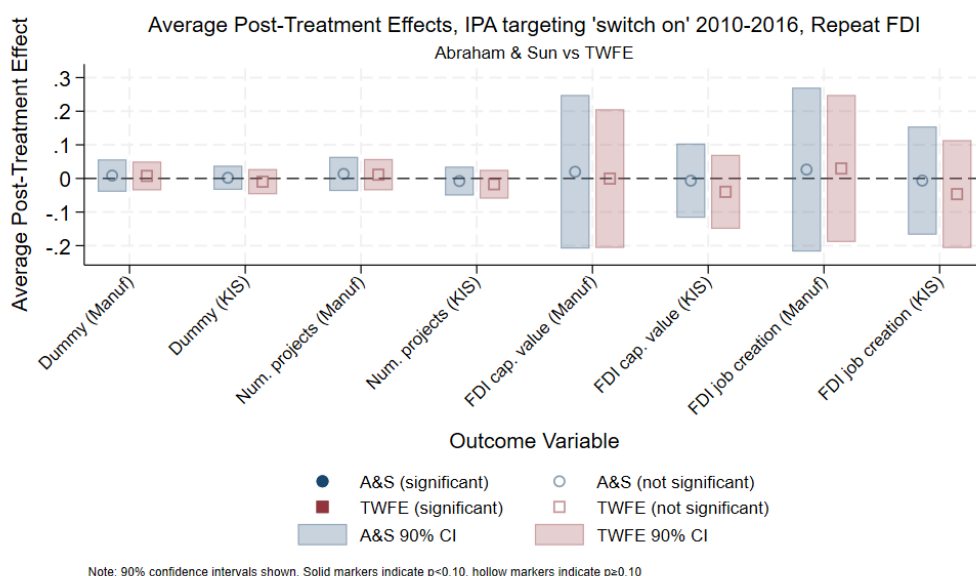
5.1 Pre-Brexit IPA Effectiveness

A natural question arising from the existing literature on IPA effectiveness is whether the positive effects observed for KIS repeat FDI are truly specific to the post-Brexit uncertainty period, or simply reflect the well-documented effectiveness of IPA targeting in normal times. If the latter, my results would represent a continuation of pre-existing IPA

impacts rather than evidence that uncertainty contexts specifically activate additional, ecosystem-building mechanisms.

Both under a standard TWFE model and when applying corrections for potential biases in staggered estimation (Sun & Abraham 2021), IPA targeting strategies show no significant association with increased inward FDI during the pre-Brexit period (2010–2016), for either manufacturing or KIS sectors (Figure 9). While this null result should be interpreted with caution given the limited post-treatment windows available, it provides reassurance that the positive effects observed for KIS repeat FDI under specifications (1)–(4) are unique to the post-Brexit uncertainty period rather than reflecting a continuation of pre-existing trends.

Figure 9: **Average post-treatment effects of IPA targeting strategies switching on between 2010 and 2016, repeat FDI.**



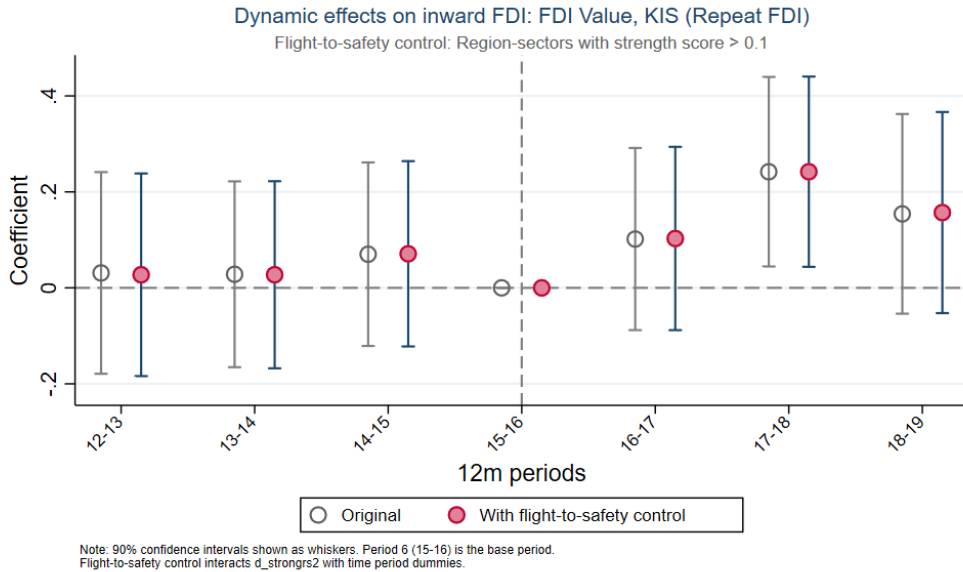
Note: The equivalent figure for new FDI is shown in Appendix Figure A3. Individual event study plots for each outcome are available upon request.

This pattern – where IPA effectiveness ‘switches on’ during the Brexit uncertainty shock – suggests that English IPAs play a distinct protective role during crises. The concentration of effects in repeat KIS FDI – which is relatively more sensitive to ecosystem-related factors in investment decision-making – suggests that ecosystem-building mechanisms, rather than traditional information-provision, drive IPA effectiveness under uncertainty.

5.2 Flight to Safety Effects

Figure 10 presents the dynamic treatment effects for repeat KIS FDI capital value when including the $Leading_{rs} \times Period_k$ control described in Section 3.1.1. The results remain substantively unchanged from the main specification ⁶. Two factors explain this stability. First, the correlation between treatment status and region-sector strength is modest ($\rho = 0.038$, $p < 0.01$): while IPAs were somewhat more likely to target stronger region-sectors, the relationship is weak, limiting the scope for flight-to-safety dynamics to confound my estimates. Second, examination of the $Leading_{rs} \times Period_k$ coefficients reveals no significant differential FDI growth in leading region-sectors during the post-Brexit period – the flight-to-safety surge that this test is designed to capture does not appear to be a prominent feature of post-Brexit FDI patterns in my data. Together, these findings suggest that while flight-to-safety is a theoretically important concern in uncertainty contexts, it does not materially bias the treatment effects estimated in this study.

Figure 10: **Dynamic treatment effects on KIS repeat FDI capital value with and without flight-to-quality control**



5.3 Pre-existing Differential Trends

A related concern is that the estimated treatment effects may reflect pre-existing differential trends between targeted and non-targeted region-sectors rather than genuine responses to Brexit uncertainty. If IPAs systematically targeted region-sectors that were already on upward FDI trajectories, the post-Brexit divergence I attribute to IPA effectiveness could simply be a continuation of these pre-existing dynamics.

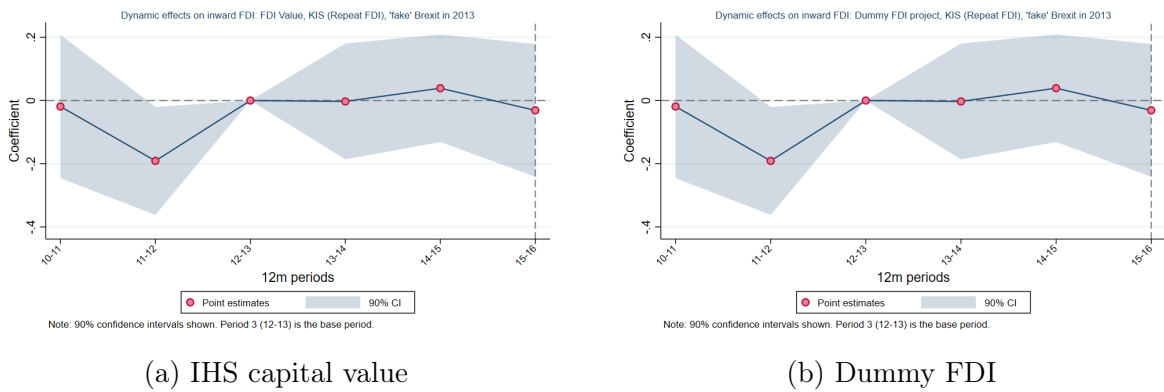
⁶Results are robust to alternative thresholds (0.05, 0.20), and available upon request.

I address this concern through two complementary approaches. First, my preferred specifications include a control for pre-Brexit FDI growth interacted with a post-Brexit indicator ($PreBrexitGrowth_{rs} \times Post_t$). This parametric adjustment allows region-sectors with stronger pre-treatment growth to follow different post-treatment trajectories, absorbing any mechanical relationship between historical momentum and post-Brexit outcomes. As reported in Table 5, the inclusion of this control does not materially attenuate the estimated treatment effects, suggesting that differential pre-existing momentum is not driving the results.

Second, I implement a placebo test that provides a more direct assessment of whether the treatment-control comparison yields spurious effects in periods when no true effect should exist. Specifically, I restrict the sample to the pre-Brexit period (2010–2016) and estimate the dynamic specification with a ‘fake’ Brexit date in 2013, using the 2012–13 period as the base. If my results were driven by pre-existing differential trends rather than a genuine response to Brexit uncertainty, I would expect to observe significant ‘effects’ emerging in the fake post-treatment periods (2013–14, 2014–15, 2015–16).

Figure 11 presents the results of this placebo test for repeat KIS FDI. The pattern is reassuring: coefficients in the fake post-treatment periods are small, statistically insignificant, and show no systematic upward drift. This contrasts sharply with the main results (Figure 5), where clear positive effects emerge in the actual post-Brexit periods (2017–18, 2018–19). The absence of spurious effects under fake treatment timing strengthens confidence that the estimated post-2016 effects represent genuine institutional responses to Brexit uncertainty rather than the continuation of pre-existing trends.

Figure 11: Placebo test: Dynamic treatment effects with fake Brexit date in 2013



Taken together, these two approaches provide convergent evidence that the main results are not artefacts of pre-existing differential trends. The parametric growth control addresses the concern within the main specification, while the placebo test demonstrates that the difference-in-differences design does not generate spurious effects when applied to periods without genuine treatment.

6 Conclusions

Uncertainty sparked by systemic shocks is known to negatively affect inward Foreign Direct Investment, and the Brexit negotiation period provides a salient example of how transition-era disruptions reshape international capital flows. The UK saw substantially lower FDI inflows in the years following the 2016 referendum (Serwicka & Tamberi 2018, Bloom et al. 2019). However, these effects were highly heterogeneous across sectors, FDI types, and regions. This paper has investigated whether the UK’s regional institutions – specifically, Investment Promotion Agencies’ targeted strategies to support particular sectoral ecosystems before Brexit was even contemplated – created conditions for MNE investors to better resist uncertainty, explaining why certain types of FDI held up better than others.

Drawing on literature from International Business and Economic Geography, I posit that MNEs operating in regional ecosystems characterised by greater density of and access to key assets – at least partly because regional institutions have targeted resources toward building these – are more likely to perceive operational benefits that motivate continued investment when faced with costly uncertainty. I find that pre-2015 IPA sector-targeting strategies were effective at increasing inward FDI specifically in repeat investments by established MNEs and in Knowledge Intensive Services (KIS) sectors. Effects concentrate on the intensive margin: targeted region-sectors attracted 18.4% higher capital value and 20.6% higher job creation compared to non-targeted region-sectors. While targeted region-sectors were also 4.1 percentage points more likely to receive any repeat KIS FDI at the extensive margin, this effect is weaker and not corroborated by my alternative extensive-margin measure (project count). This pattern suggests that IPA targeting primarily influenced the *scale* of existing investors’ commitments rather than simply determining whether investment occurred at all – consistent with established MNEs leveraging deeper access to regional ecosystem assets to expand operations, rather than IPAs merely retaining investor presence. By contrast, new FDI and manufacturing sector investments saw no significant impact. Taken together, this pattern indicates that ecosystem-building, rather than information provision, drives IPA effectiveness during fundamental uncertainty: established investors already embedded in regional ecosystems respond by deepening their commitments, while new investors for whom information assistance would be most valuable – and manufacturing investors facing trade-related uncertainty that no amount of information can resolve – do not.

My analysis reveals meaningful heterogeneity in these effects across both geography and institutional capacity that further corroborates this interpretation. The protective effect of IPA targeting concentrates outside South East England, where regional ecosystem assets are valuable but less immediately accessible to external investors, making

active institutional brokering most consequential. Within the economically prosperous South East core, where firms already have ready access to dense networks and specialised resources, IPA-facilitated ecosystem access appears redundant. Equally striking is the concentration of effects among Combined Authority IPAs – statutory bodies with devolved powers and substantially greater resources than Local Enterprise Partnerships or Upper Tier Local Authorities. This suggests that ecosystem-building during uncertainty requires the kind of institutional capacity that enables sustained coordination across universities, industry associations, skills providers and infrastructure bodies: precisely the convening power that Combined Authorities possess and that fragmented LEPs and UT-LAs lack. More broadly, these findings illustrate how place-based institutions mediate the spatial consequences of global transition shocks, producing uneven but institutionally contingent regional outcomes.

This paper contributes to our understanding of how regional institutions can support the broader aims of regions with respect to internationalisation – attracting quality, durable investments that foster regional resilience to shocks. The findings carry implications for regional development policy in an era of structural transformation. They suggest that strengthening ecosystem-oriented institutional capacity may be particularly important in regions lacking dense agglomeration advantages, where resilience to global shocks depends more heavily on active coordination and asset articulation. The importance of institutional capacity – illustrated here by the contrast between Combined Authorities and LEPs – further implies that the form of regional governance matters, not just its presence. IPAs offer one institutional vehicle for this work, but the broader lesson extends to regional development agencies, local authorities, and other place-based institutions charged with supporting economic resilience.

Several avenues for future research remain open. More work is needed to disentangle exactly how regional institutions deliver ecosystem benefits, including the nature of coordination with other regional and institutional actors. Further attention should focus on how the heterogeneous and often limited institutional capabilities of British regional IPAs – particularly following the dissolution of England’s Regional Development Agencies in 2011 – may have shaped their ability to engage with MNEs during periods of disruption. Finally, future work might explore whether these results generalise to other uncertainty contexts beyond Brexit, and whether institutional capabilities developed during one crisis enhance resilience to subsequent shocks – a question of increasing relevance in an era of compounding global disruptions.

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